KAHBOD AEINI

Waterloo, Ontario, Canada
kaeini@uwaterloo.ca — LinkedIn — GitHub — kahbod.org — Google Scholar

SUMMARY

Software engineer and distributed systems researcher. Designs and ships GraphQL and REST APIs, SQL-backed services, and application security features for a production retail energy platform, while completing a PhD on concurrency and persistent memory at the University of Waterloo. Strong across the backend stack — C#/.NET and Python services, heavy SQL, relational and document databases, Docker and Linux — plus deep expertise in concurrent and lock-free data structures on real multicore hardware. Best Paper Award at PODC 2026 for work on wait-free synchronization. Carries a full-time PhD and a production engineering role in parallel, and has shaped hiring and onboarding and led teams in both.

TECHNICAL SKILLS

Languages: C#, SQL, GraphQL, Python, C/C++, Java, R, Haskell, Prolog
APIs & Backend: GraphQL schema & resolver design, REST API design, .NET / ASP.NET Core, Entity Framework, Django, EDI integrations, Docker, Git, Linux / NixOS
Security: authentication & authorization, role- and permission-based access control, encryption & secure data exchange, secure API design
Databases & Data: Microsoft SQL Server, PostgreSQL, MySQL, MongoDB, complex SQL & query optimization, schema design, ETL pipeline design, Pandas, NumPy
Concurrency & Systems: multicore & NUMA programming, Intel RTM / transactional memory, OpenMP, atomic primitives (CAS), lock-free & wait-free data structures, epoch-based memory reclamation, `perf` profiling
Machine Learning: PyTorch, TensorFlow, scikit-learn, OpenCV, feature engineering, model evaluation

EXPERIENCE

Software Engineer Dec 2025 – Present
ArcTrade (part-time, alongside PhD)

- Design and build GraphQL and REST APIs for ArcTrade's retail energy operating system (Retail eOS), used by retail electricity providers in North American markets.
- Core contributor to the platform's security module: authentication, authorization, and permission-based access control governing which users and partners may read or act on sensitive market and financial data.
- Write and optimize substantial SQL against the platform's relational data model, backing API resolvers and reporting paths with correct, performant queries.
- Implement secure, automated electronic data interchange (EDI) with external trading partners, enabling encrypted and verifiable exchange of business transactions.
- Contribute to components supporting energy trading and risk management (ETRM), pricing, billing, and settlement workflows, where correctness of financial calculations is non-negotiable.
- Helped design the team's hiring and onboarding process for new engineers.

Data Engineer Feb 2022 – Sep 2022
AI Lida

- Developed, maintained, and optimized ETL data pipelines in Python and PostgreSQL serving analytics workloads.
- Partnered with the security department to gather requirements and deliver data solutions aligned with business objectives.
- Interviewed, hired, trained, and supervised a team of 5 data engineering interns, running onboarding and performance feedback.

RESEARCH EXPERIENCE

Graduate Research Assistant Jan 2026 – Present
Distributed Algorithms and Systems Lab, University of Waterloo — Advisor: Prof. Wojciech Golab

- Design and analyze shared-memory algorithms and synchronization primitives for systems with persistent (non-volatile) memory.
- Focus on recoverable objects: concurrent data structures whose state stays consistent across crash failures — directly relevant to durable, fault-tolerant storage systems.

Graduate Research Assistant 2023 – 2025
University of Calgary — Supervisor: Prof. Philipp Woelfel

- Designed the first constant-space randomized wait-free locking algorithm for mutual exclusion in shared-memory systems; proved correctness and wait-freedom, and analyzed performance with probability theory.
- Resulting paper received the **Best Paper Award** at PODC 2026, the flagship conference in distributed computing.

Undergraduate Research Assistant (Remote) Feb 2023 – Aug 2023
Johns Hopkins University — Supervisor: Prof. Mehdi Pirooznia

- Built machine learning pipelines and feature selection workflows in Python/R to predict immune cell composition in lung cancer; ran differential expression and enrichment analyses and automated reporting for reproducible results.

SELECTED PROJECTS

Concurrent Data Structures on Multicore Hardware 2026
Implemented and benchmarked TLE-based concurrent binary search trees and expandable probing hash tables using Intel Restricted Transactional Memory on a 192-thread NUMA server. Analyzed commit/abort behavior, false sharing, NUMA effects, and epoch-based memory reclamation with `perf`.

Secure Chat Application GitHub
Built a Python chat application in a team of three, hardened with multiple computer-security mechanisms: authentication, encrypted transport, and integrity checks.

EDUCATION

University of Waterloo Jan 2026 – Present
PhD, Electrical and Computer Engineering — Advisor: Prof. Wojciech Golab

University of Calgary 2023 – 2025
MSc, Computer Science — Thesis: *A Space-Optimal Randomized Wait-Free Lock*

Aryamehr (Sharif) University of Technology 2019 – 2023
BSc, Computer Engineering — Thesis: *Fraud Detection from Sequences of Bank Transactions*

AWARDS

Best Paper Award, 45th ACM Symposium on Principles of Distributed Computing (PODC) 2026
Departmental Research Award, Department of Computer Science, University of Calgary 2023
Top 100 Nationally (94th of ∼160,000, Entrance Exam) & **National Elites Fellowship**, Iran 2019

SELECTED PUBLICATIONS

Recoverable Failure-Atomic Lock-Free Double-Width-Compare-And-Swap from Double-Width-Compare-And-Swap — under review, OPODIS 2026.
A Faster Universal Construction of Detectable Read-Modify-Write Objects for Persistent Memory — under review, PPoPP 2027.
Simple and Efficient Randomized Wait-Free Locks — PODC 2026. **Best Paper Award.**
Analyzing Linearizability in Relativistic Distributed Systems — ApPLIED Workshop @ PODC 2026.

LEADERSHIP & MENTORING

Head Teaching Assistant, University of Calgary (2023–2026) and Sharif University of Technology (2020–2023). Led TA teams of up to 25 across algorithms, data structures, and theoretical computer science, supporting 400+ students; owned assignment design, tutorial delivery, and grading standards.